Decomposition of Concentration-Index Trees

Moses Mburu

1 Results

1.1 Exploratory Data Analysis

In the DRC Data Set(The DRC Program Program, 2007), 10.32% of children died before reaching age five (Figure 1). Deaths were more commonly observed among children from disadvantaged or rural backgrounds. The children who died differed descriptively from those who survived. Low-wealth households accounted for a larger share of deaths than of survivors (51.3% versus 42.8%). The same pattern was present for rural residence (69.3% versus 60.7%), no skilled attendance at birth (65.6% versus 56.8%), and mothers with no education (67.5% versus 55.2%) (Table 2). Mortality also differed across provinces, ranging from 6.2% in Kinshasa to 14.2% in Katanga (Figure 12). Next we use the direct and indirect methods for decomposition.

Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 Decomposition of determinants of under-five mortality

1.2.1 Indirect methods comparison (regression and predictive-forest SHAP)

From the indirect decomposition, we see that the largest regression contribution came from mother's education: 23.1% under CI and 28.7% under CIg/CIc (Table 1). Father's occupation in agriculture was second, contributing 17.9% under CI and 19.9% under CIg/CIc. In the predictive-forest SHAP decomposition, the contributions were less concentrated in one or two variables.

Residence and household wealth each contributed about 12% under CI, CIg and CIc, followed by mother's occupation, father's occupation, mother's education and skilled birth attendance. The fitted regression model is reported in Table 3. Details of the fitted predictive forest are provided in Table 4 and Table 5, and its SHAP contributions are shown in Figure 13.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion.

	Regression			**Predictive-forest SHAP**			
Variable	**CI (%)**	**CIg (%)**	**CIc (%)**	**CI (%)**	**CIg (%)**	**CIc (%)**	**L (%)**
Mother's education	23.058	28.726	28.726	11.110	11.110	11.110	12.684
Father's occupation: Agriculture	17.863	19.885	19.885	4.619	4.619	4.619	5.060
Mother's occupation: Agriculture	11.684	14.393	14.393	3.474	3.474	3.474	3.262
Household wealth	9.116	8.788	8.788	11.531	11.531	11.531	6.980
Type of residence	8.733	11.866	11.866	21.630	21.630	21.630	21.038
Skilled birth attendance	7.848	9.988	9.988	10.058	10.058	10.058	7.107
Region: Equateur	3.761	1.066	1.066	3.165	3.165	3.165	4.165
Father's occupation: Household, unskilled, not working	3.370	0.955	0.955	-0.430	-0.430	-0.430	-0.343
Region: Kinshasa	3.221	0.647	0.647	1.246	1.246	1.246	2.054
Region: Sud-Kivu	2.006	0.213	0.213	-0.688	-0.688	-0.688	-0.434
Region: Kasai Oriental	1.746	0.513	0.513	0.106	0.106	0.106	0.096
Father's education	1.462	0.869	0.869	3.043	3.043	3.043	3.500
Region: Maniema	1.075	0.083	0.083	3.124	3.124	3.124	4.218
Region: Orientale	0.984	0.251	0.251	0.283	0.283	0.283	0.107
Region: Bas-Congo	0.678	0.060	0.060	0.048	0.048	0.048	0.474
Mother's occupation: Household, unskilled, not working	0.599	0.306	0.306	0.094	0.094	0.094	-0.068
Birth order and interval: 5+ short interval	0.528	0.118	0.118	1.214	1.214	1.214	4.518
Region: Kasai Occidental	0.490	0.115	0.115	0.007	0.007	0.007	0.102
Birth order and interval: 5+ long interval	0.437	0.253	0.253	-0.333	-0.333	-0.333	-0.040
Sex of the child	0.418	0.454	0.454	0.068	0.068	0.068	-0.005
Region: Katanga	0.296	0.067	0.067	-0.155	-0.155	-0.155	-2.068
Mother's age at birth	0.261	0.249	0.249	0.505	0.505	0.505	0.613
Region: Nord-Kivu	0.208	0.018	0.018	0.619	0.619	0.619	0.429
Birth order and interval: 2-4 long interval	0.154	0.115	0.115	0.409	0.409	0.409	0.106
Birth order and interval: 2-4 short interval	0.004	0.001	0.001	0.458	0.458	0.458	0.231

1.3 Direct methods (concentration-index trees and forests)

1.3.1 Rank Dependent Concentration-Index Tree

The Corrected Concentration-Index Tree CIc

We start with a tree based on the corrected concentration index, appropriate for bounded outcomes(Erreygers, 2009). In this method, the tree shows that inequality in under-five mortality is structured primarily by maternal education, father's agricultural occupation, household wealth, skilled birth attendance, and residence in Katanga Figure 2. The lowest-mortality terminal group consists of children whose mothers had some education and whose fathers were not in agriculture, with an under-five mortality rate of about 6.5%. At the opposite end, children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga had the highest mortality rate, about 17.8%. Thus, the corrected rank-dependent tree identifies a clear socioeconomic gradient, but one that is expressed through combinations of education, occupation, household wealth, delivery care, and region rather than through wealth alone.

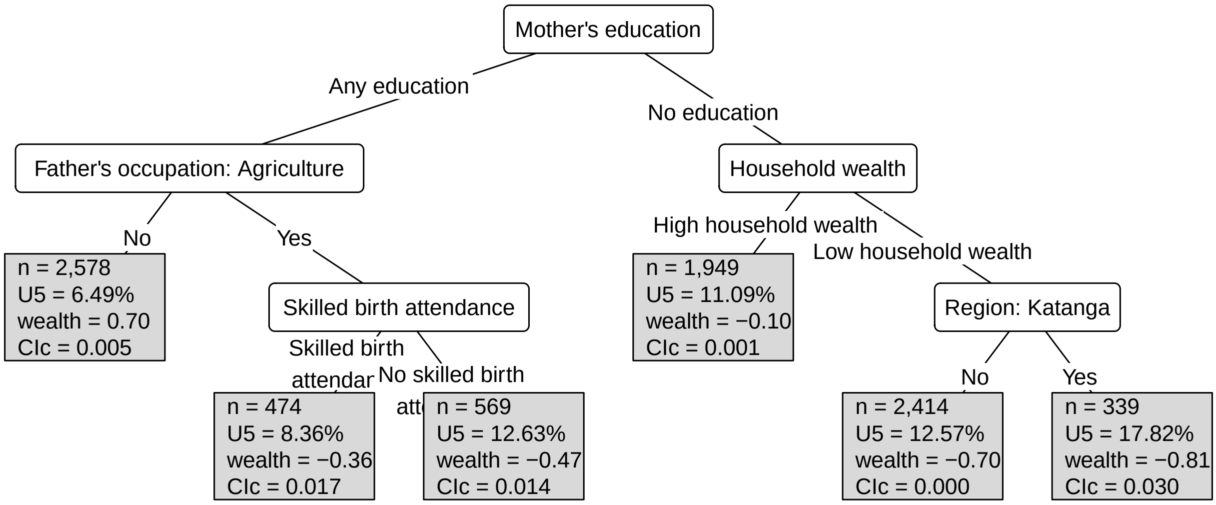

Figure 2: Concentration-index tree fitted with the CIc criterion.

Generalised and standard concentration-index trees CIg and CI

The appendix also shows the trees fitted using the generalised and standard concentration indices. The generalised concentration-index tree is almost identical to the corrected concentration-index tree Figure 15. It again separates children first by maternal education and father's agricultural occupation, then by skilled birth attendance, household wealth, and residence in Katanga. The mortality pattern is therefore substantively unchanged: the lowest-risk group consists of children whose mothers had some education and whose fathers were not in agriculture, while the highest-risk group consists of children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga, with mortality around 17.8% Figure 15. The standard concentration-index tree gives a similar structure. However, it adds an extra distinction among children whose mothers had some education and whose fathers were not in agriculture, splitting them by short preceding birth interval Figure 16. Children in this otherwise advantaged group without a short birth interval have the lowest mortality, about 6.0%. In contrast, those with a short birth interval have higher mortality, about 11.7% Figure 16. Apart from this additional demographic split, the standard CI tree points to the same main sources of inequality as the corrected and generalised trees: maternal education, father's agricultural occupation, delivery care, household wealth, and residence in Katanga.

1.3.2 Level dependent Concentration-Index Tree L

We then fit a tree based on the level-dependent concentration index (Erreygers and Kessels, 2017) criterion, L. In this method, the selected tree shows that inequality in under-five mortality is structured primarily by rural residence, father's agricultural occupation, household wealth, and province, especially Kinshasa, Equateur, and Katanga Figure 3. The lowest-mortality terminal group consists of urban children in Kinshasa whose fathers were not in agriculture, with an under-five mortality rate of about 5.1%. The highest-mortality terminal group consists of rural children living in low-wealth households in Katanga, with an under-five mortality rate of about 18.5%. Thus, the level-dependent tree tells a more spatial story of inequality: mortality risk is first organised by the rural-urban divide, then by occupation, wealth, and region. This shows that the level-dependent method found a different structure compared to the rank-dependent methods

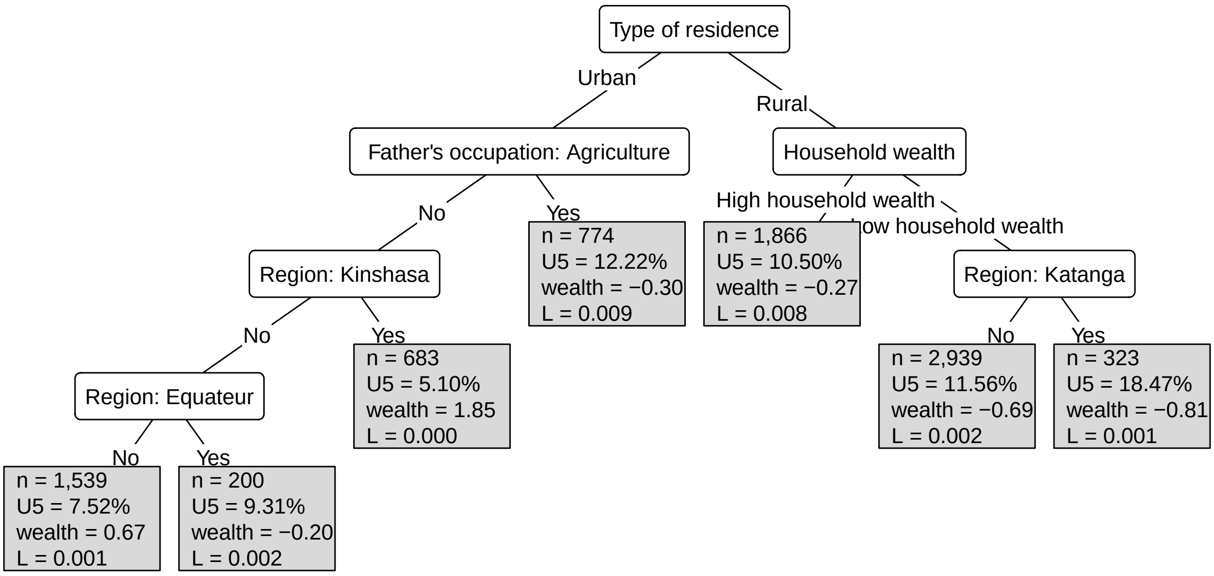

Figure 3: Level dependent concentration-index tree for under-five mortality in the DRC.

1.3.3 Tree validation plots

The rank-dependent trees performed well on the training data but poorly on the validation data. The CI, CIc, and CIg trees removed approximately 87 to 88% of root-node inequality during training, but had negative validation gains at the selected settings: -75.5% for CI and -73.1% for both CIc and CIg Figure 4, Table 7. In this criterion, a negative validation gain means that the sum of the terminal-node concentration-index values in the validation fold exceeded the concentration-index value at the root node. Thus, although these trees identified substantial rank-based structure in the fitted sample, the same partitions did not reduce inequality when applied to held-out folds. Unlike rank-dependent trees, the level-dependent tree maintained a positive validation gain throughout the tuning path. At the selected setting, it removed 70.5% of root-node inequality in held-out folds while retaining an average of 6.8 terminal nodes Figure 4,

Table 7. The rural-urban, occupational, wealth, and regional partition identified by the L tree was therefore more stable under the validation criterion used in this thesis.

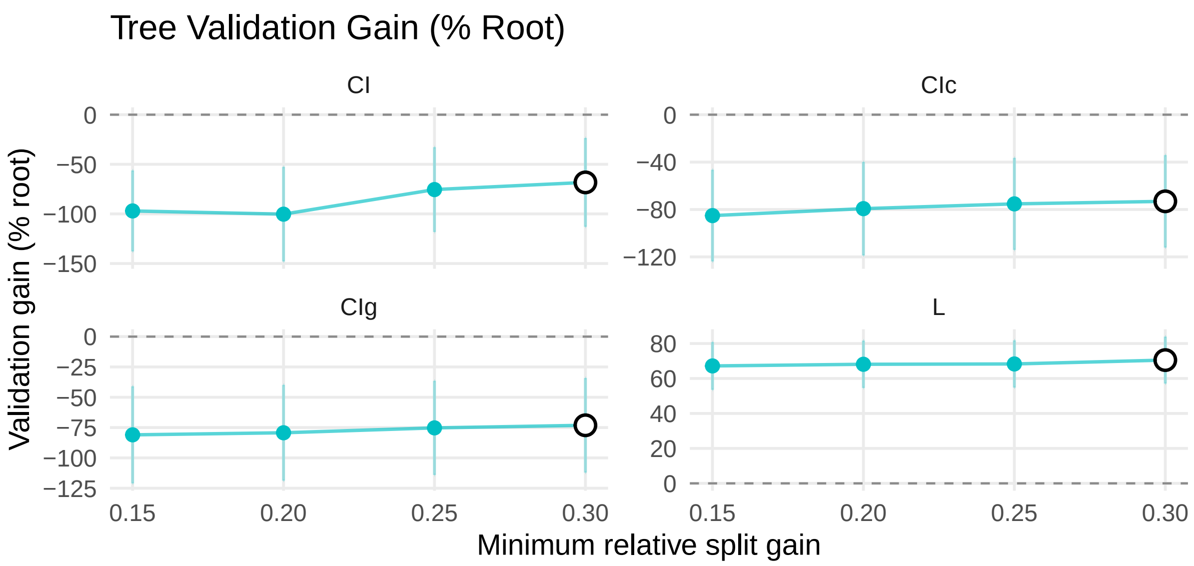

Figure 4: Tree validation gain and complexity by minimum relative split gain.

Tree complexity plots

Tree complexity decreased as the minimum relative split gain increased, since larger thresholds allowed only splits that removed a larger share of parent-node impurity Figure 5. The selected trees remained small, with an average of 7.5 terminal nodes for CI, 7.3 for CIc and CIg, and 6.8 for L Table 8, Figure 5.

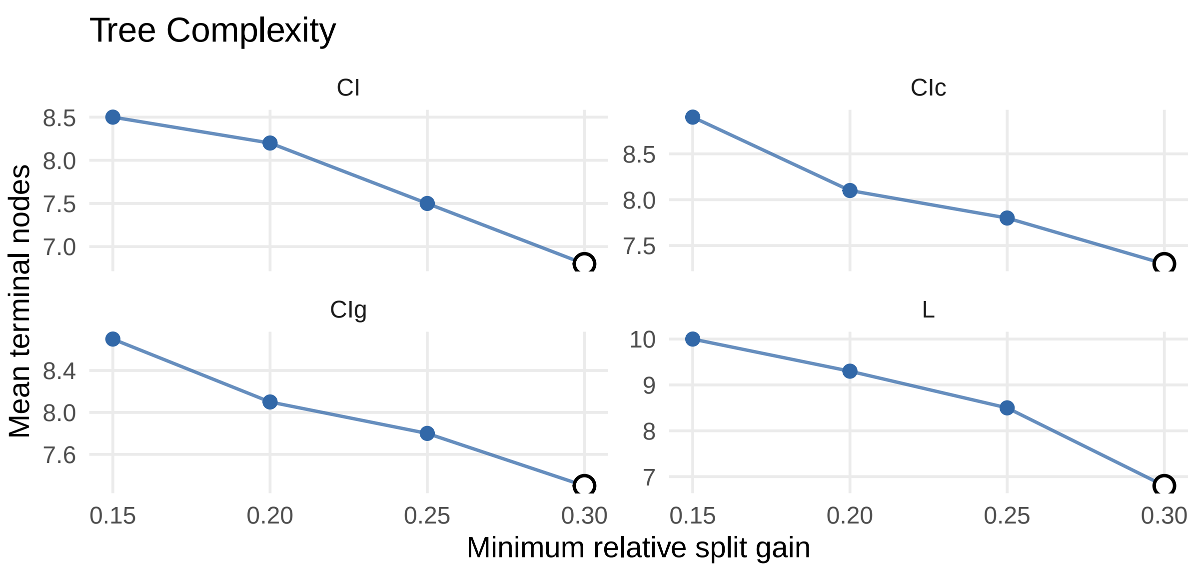

Figure 5: Tree complexity by number of terminal nodes.

Rpart comparison

The rpart comparison tree recovered the same upper-level structure as the main CI tree (Figure 16; Figure 6). Both trees first split on maternal education, then separated children by father's agricultural occupation among those whose mothers had some education, and by household wealth among those whose mothers had no education. The high-mortality branch was also similar, identifying children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. The main difference was in the lower branches: the rpart comparison introduced a split by long preceding birth interval, while the main CI tree used a short preceding birth-interval split. This suggests that the main rank-dependent structure was not driven only by the custom implementation, although some lower-level refinements depended on the tree-growing and stopping rules.

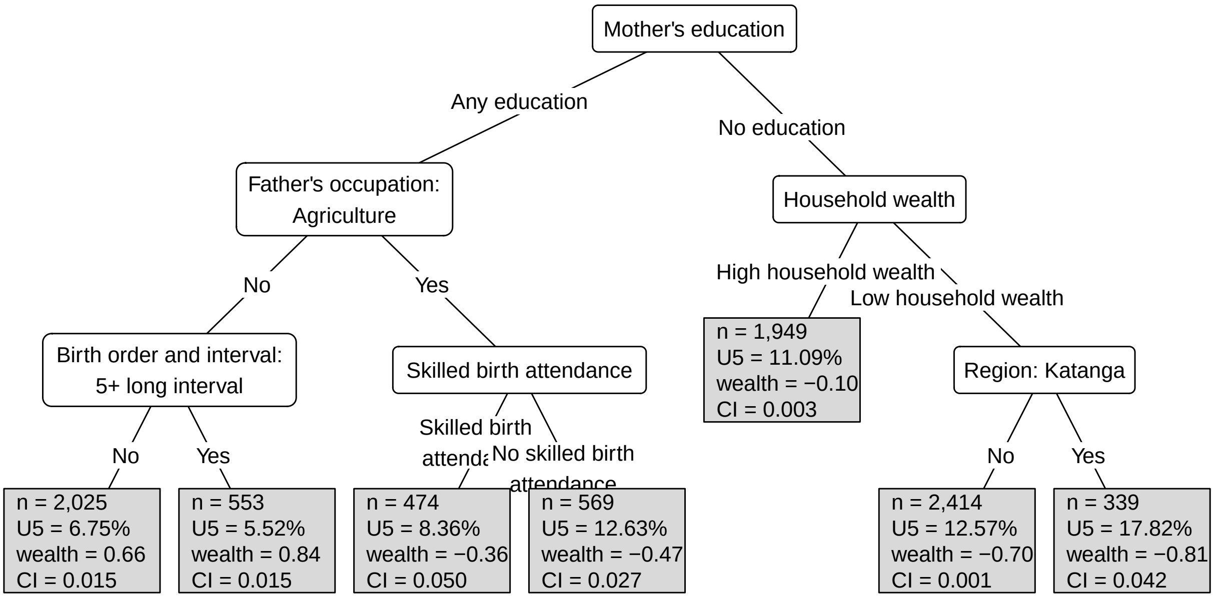

Figure 6: Rpart comparison tree using CI impurity.

1.3.4 Concentration index forest and surrogate tree

Forest validation plots

We evaluated concentration-index forests using the same validation criterion as the single trees. For the forests, the minimum relative split gain was tuned to 0.35, since larger values produced near-root-only forests with mean terminal-node counts close to 1. At that point, further validation failed to reveal a meaningful structure. For the rank-dependent criteria, the forests reduced overfitting but did not stabilise the partitions. The CI, CIc, and CIg forests removed 29.6%, 38.5%, and 38.7% of root-node inequality in training, compared with about 87 to 88% for the corresponding single trees Table 10, Table 7. Their validation gains remained negative: -25.0% for CI, -28.1% for CIc, and -34.7% for CIg Figure 7, Table 10. Thus, averaging across trees reduced the validation loss, but did not produce a reproducible rank-dependent subgroup structure. The tuning path shows the trade-off. Validation gain improved as the minimum relative split gain increased, but the component trees also became shallower Figure 7, Figure 18. At the selected setting, the mean terminal-node counts were 2.4 for CI, 2.9 for CIc, and 2.9 for CIg Table 11. The level-dependent forest remained more stable. It removed 95.1% of root-node inequality in training and 69.7% in validation Table 10. The selected L forest had an average of 4.1 terminal nodes per tree, compared with 6.8 for the selected L tree Table 11, Table 8.

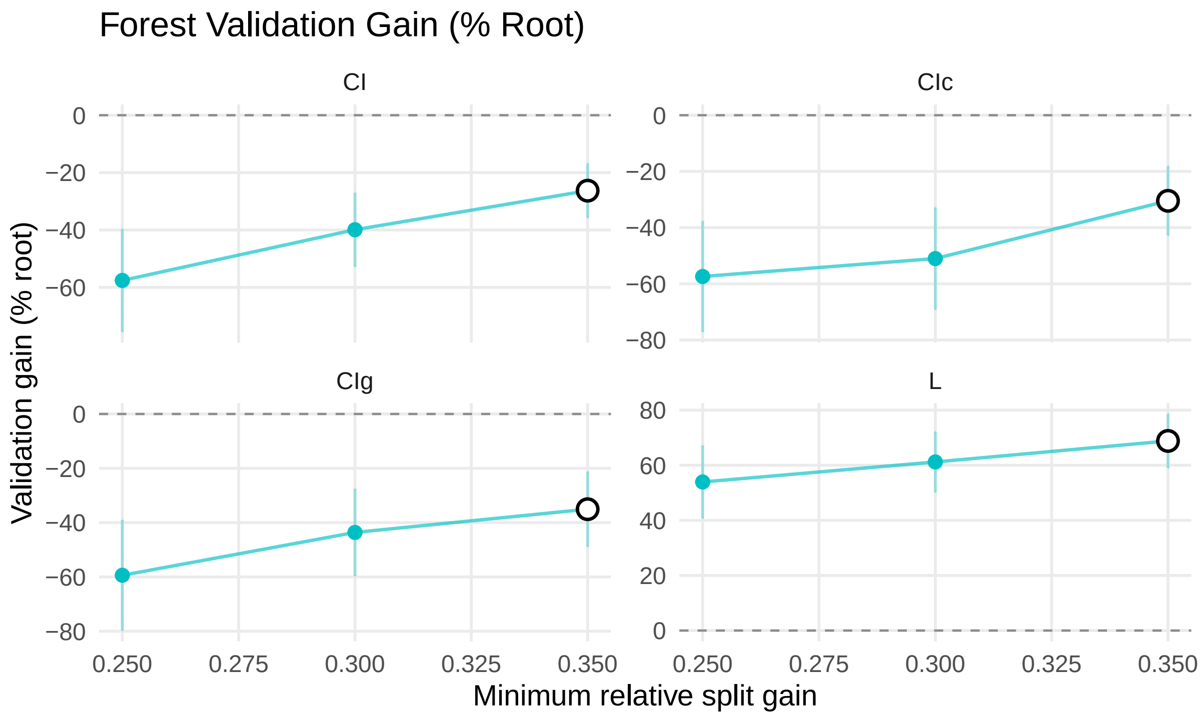

Figure 7: Forest validation gain and complexity by minimum relative split gain.

Surrogate trees from Forest Predictions

To interpret the selected forest, we fitted a surrogate tree to summarise its predictions (Figure 8). The surrogate tree retained the main structure seen in the level-dependent tree.

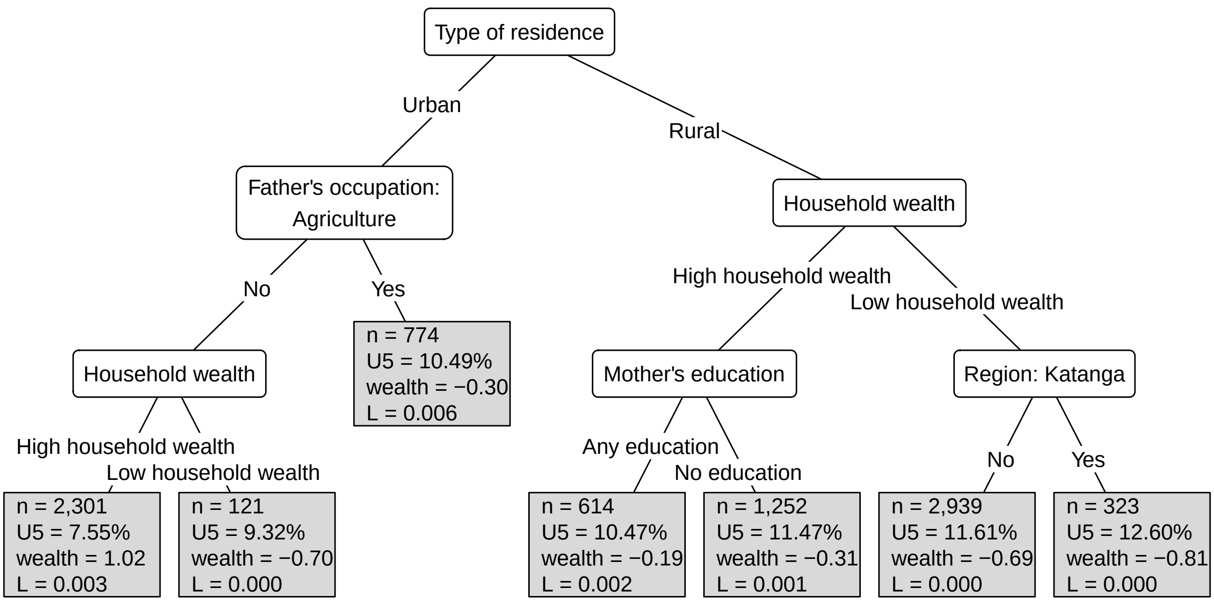

Figure 8: Surrogate tree summarising the selected concentration-index forest predictions.

The first split was type of residence, separating urban and rural children. Among urban children, the main divisions were father's agricultural occupation and household wealth. Among rural children, the main divisions were household wealth, maternal education, and residence in Katanga. The lowest terminal-node mortality was observed among urban children in the high-wealth branch, while the highest terminal-node mortality was observed among rural, low-wealth children in Katanga. Compared with the single L tree, the surrogate gives a smoother version of the same rural-urban, socioeconomic, and regional pattern, with less extreme terminal-node mortality differences. The criterion-specific forest surrogate trees are shown in the appendix for comparison (Figure 21; Figure 22; Figure 20).

1.3.5 Variable importance

The variable-importance plots support the interpretation of the selected level-dependent models. In both the L tree and L forest, the type of residence gives the largest reduction in concentration-index impurity Figure 9. In the single tree, residence is followed by father's agricultural occupation, Kinshasa, household wealth, Equateur, and Katanga, reflecting a structure organised around the rural-urban divide, occupational position, household wealth, and regional differences.

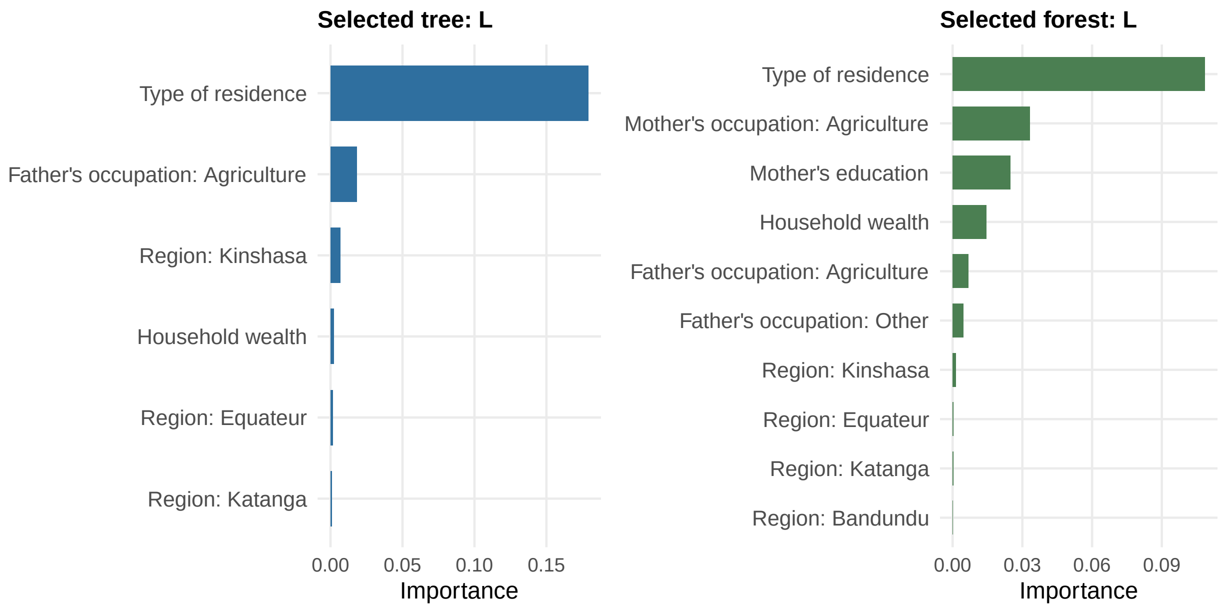

Figure 9: Variable importance for the selected concentration-index tree and forest.

The L forest gives a less concentrated ranking than the single tree. Residence is still the first variable, but the forest also gives weight to mother's agricultural occupation, maternal education, household wealth, father's agricultural occupation, and regional indicators. In the appendix plots, the same groups of variables appear again under the other criteria: residence, region, occupation, education, and wealth Figure 14, Figure 19.

1.4 Simulation Study

1.4.1 Null case

From (Figure 10) we see that the algorithm developed here did not find a meaningful structure and behaved as expected. This shows that the tree-growing rule does not create subgroup structure when the outcome and socioeconomic ranking are independent, since the tree uses concentraion index as an impurity measure, which measures how sociaeconomic ranking and outcome covary.

(a) CI (b) CIg (c) CIc (d) L

Figure 10: Null simulation concentration-index trees by criterion.

1.4.2 Poor subgroup case

In the positive-control simulation, the method(the concentration index tree with CIc as an impurity measure) recovered the planted subgroup structure. The tree first separated urban from rural observations, then split the rural group by province and maternal education (Figure 11). This structure matched the way the simulated outcome was generated: poorer observations in the planted rural, low-education, selected-province subgroup had higher outcome prevalence. The terminal nodes also followed the expected gradient, with the highest prevalence among observations in the most disadvantaged planted subgroup.

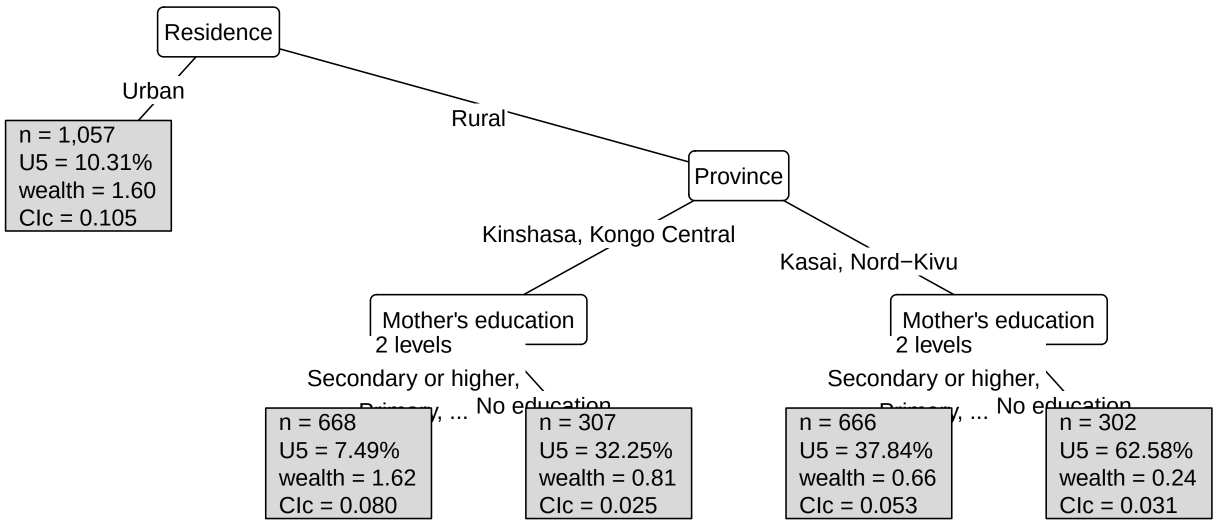

Figure 11: CIc concentration-index tree recovered from simulated data with a known subgroup-generating mechanism.

We see the same pattern across the sensitivity plots in the appendix the trees developed using CI, CIg, and CIc, and L as impurity measures(Figure 24; Figure 25; Figure 26).

2 Discussion

2.1 Main findings

This thesis developed and applied a tree-based model for analysing socioeconomic inequality in under-five mortality using the concentration index impurity. The main finding is that inequality in the DRC data could not be explained by household wealth alone. Under-five mortality was also related to residence, maternal education, skilled birth attendance, parental occupation, and province. Overall, 10.32% of children died before age five. The children who died were more often from low-wealth households, rural households, mothers with no education, births without skilled attendance, and some provinces, especially Katanga. This is in line with earlier DRC work on geographic variation in under-five mortality, and with decomposition studies showing that wealth, education, residence, and occupation contribute to mortality inequalities in African settings (Van Malderen, Van Oyen and Speybroeck, 2013; Kandala *et al.*, 2014; Van Malderen *et al.*, 2019).

The indirect methods provided a variable view of this inequality. In the regression decomposition, mother's education made the largest contribution, accounting for 23.1% under CI and 28.7% under CIg and CIc. Father's agricultural occupation was the second-largest contributor, accounting for 17.9% under CI and 19.9% under CIg and CIc. These results are consistent with earlier decomposition work, which found that household wealth, parental education, occupation, and residence were important contributors to under-five mortality inequality (Van Malderen, Van Oyen and Speybroeck, 2013; Van Malderen *et al.*, 2019). The SHAP analysis (from the fitted random forest) gave a less concentrated picture: residence and household wealth each contributed about 12%, followed by mother's occupation, father's occupation, mother's education, and skilled birth attendance. Shap-based decomposition provides an addition to the regression methods by allowing the use of more flexible models such as Random Forests or XGBoost.

The direct concentration-index trees added the subgroup interpretation that is less visible in the regression-based decomposition. While the indirect methods ranked variables by their contribution to inequality, the trees showed how these variables combined within the data to define groups with different inequality patterns. The rank-dependent trees based on CI, CIg, and CIc identified a structure involving maternal education, father's agricultural occupation, skilled birth attendance, household wealth, and Katanga. Mortality was lower among children whose mothers had some education and whose fathers were not in agriculture, and higher among children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. However, these partitions did not validate well. Although the rank-based methods removed about 87% to 88% of root-node inequality in training, their validation gains were negative, suggesting that the strong training structure did not transfer well to held-out folds.

2.2 Rank-dependent and level-dependent views of inequality

From these analyses, we see differences in the results (Figure 16; Figure 15; Figure 2) in subgroups formed by the two ways of defining the impurity measure when developing the concentration index tree. In the rank-dependent criteria CI, CIg, and CIc, a child's contribution depends on the child's position in the current node's socioeconomic ranking. After each split, the children in the new subgroup are ranked again, so the same child can receive a different rank because the comparison group has changed. In the level-dependent criterion, L, the child's observed socioeconomic level is carried into the subgroup, so a child's rank also reflects their wealth level.

The differences also showed up in cross-validation; the level-dependent method outperformed the rank-dependent methods. The rank-dependent trees had strong training performance but negative validation gains, while the level-dependent tree kept a positive validation gain (Figure 4; Table 7). The same pattern was seen in the forests: averaging trees reduced the validation loss for the rank-dependent criteria, but their validation gains remained below zero, whereas the L forest retained a positive validation gain (Figure 7; Table 10). The validation results show that CI, CIg, and CIc identified detailed in-sample subgroup rules, whereas L produced a more reproducible subgroup structure in this application.

These results are also mirrored by the subgroup-decomposability argument in (Erreygers *et al.*, 2017). In their comparison of rank-dependent and level-dependent measures, they show that the level-dependent index L is subgroup-decomposable and rank-dependent measures are more difficult to decompose cleanly across population groups. Decomposability of the level-dependent concentration index shown by (Erreygers *et al.*, 2017) can also explain why L performed better as a tree impurity criterion in this application: children inherit their socioeconomic level within each subgroup, rather than being re-weighted only by a new within-node rank. Although the DHS wealth index is an asset-based score(The DHS Program, 2007) and (Erreygers and Kessels, 2017) recommends using Level depedent concentration index(L) with a wealth-ranking variable that has ratio properties. The cross-validation results show that the L concentration index impurity measure yielded a structure transferable to held-out samples using the DHS wealth score.

2.3 Methodological and data limitations

The analysis is descriptive and should not be interpreted as a causal design. The DHS data are cross-sectional, and the fitted trees identify subgroup patterns rather than intervention effects. A terminal node with high mortality does not show that the splitting variables caused the outcome; it shows that the observed covariates define a group with a different mortality and inequality profile.

The choice of socioeconomic ranking variable is also a limitation. The DHS wealth index is appropriate for ordering households from poorer to richer, but it is not equivalent to measured income, consumption, or expenditure. This matters especially for the level-dependent index, because level-dependent weights are easier to justify when socioeconomic levels have a clear ratio-scale interpretation. The validation results favour the L criterion in this application, but that empirical stability should be read alongside this measurement caveat.

The SHAP analysis is also approximate. It is based on a fitted predictive forest and a sampled SHAP calculation, so the reported percentages describe how the fitted model allocates prediction differences across variables. They do not provide causal effects, and they are not the same object as the concentration-index tree impurity. Their value in this thesis is comparative: they show whether a flexible predictive model points to the same determinant groups as the direct inequality trees.

2.4 Ethical, societal, and stakeholder relevance

The main ethical issue in this thesis is not the tree algorithm itself, but the governance of the DHS data used to fit it. The analysis uses secondary survey data, so the first ethical check is how the data were collected, whether participants gave informed consent, whether confidentiality protections were applied, and whether the data are used in line with the conditions under which access was granted. The DHS Program states that "participation is voluntary" and that the respondent's "identity and information will be kept strictly confidential" (The DHS Program, 2026). The analysis should therefore be reported as a secondary analysis of protected survey data, with no attempt to identify individuals, households, or small local groups.

The second ethical issue is data protection. Even when DHS files are distributed for research in de-identified form, the analysis still uses sensitive health, household, and socioeconomic information. Data handling should therefore be checked against the data-access agreement and, where relevant, GDPR principles of lawful processing, data minimisation, purpose limitation, secure storage, and avoidance of re-identification (European Parliament and Council of the European Union, 2016). The subgroup rules should also be used carefully. They can help public-health planners see where socioeconomic inequality in under-five mortality is concentrated, but they should not be used to stigmatise provinces, rural households, or families with particular socioeconomic characteristics.

The stakeholder value of the tree approach is that the results can be explained without requiring advanced statistical knowledge. A tree diagram works like a flow chart: the reader begins at the first branch, follows the observed characteristics, and arrives at a high-risk or low-risk subgroup. This is useful for Ministry of Health officials and programme teams because the first branch immediately shows the main split in the population, while later branches show how risk differs within those groups. Regression coefficients and SHAP contributions are useful for technical comparison, but the tree and surrogate outputs translate the same inequality pattern into readable decision rules that can be discussed with non-specialist stakeholders.

3 Future research

Future research should examine why the rank-dependent criteria overfit in this application and whether alternative validation rules, bootstrap stability checks, or constraints on early splits improve their held-out performance. A second direction is to repeat the level-dependent analysis with socioeconomic variables that have clearer level meaning, such as consumption or income, where available. The method should also be applied to another DHS country sample, such as Kenya, to assess whether the same rural-urban, occupational, educational, and regional structure appears outside the DRC data.

4 Conclusion

The thesis shows that concentration-index trees can turn socioeconomic inequality in under-five mortality into interpretable subgroup rules. In the DRC application, the indirect regression decomposition highlighted mother's education and father's agricultural occupation, while the SHAP, tree, forest, and surrogate results repeatedly pointed to residence, household wealth, education, occupation, skilled birth attendance, and province. The rank-dependent trees recovered a clear in-sample socioeconomic gradient, but they did not validate well. The level-dependent tree and forest produced a more stable subgroup structure centred on rural-urban residence, occupation, wealth, and region. The method therefore adds a useful descriptive tool to classical decomposition: it does not only rank determinants, but shows how determinants combine to define inequality-relevant groups.

5 Appendix

5.1 Results

5.1.1 Appendix R1: Exploratory Data Outputs

Table 2: Weighted distribution of analysis variables by under-five mortality status

	Under-five mortality		Overall
Variable	**Alive at age 5**	**Died before age 5**	**Overall**
	(weighted N=7,464)	(weighted N=859)	(weighted N=8,323)
Household wealth			
High household wealth	4,266.9 (57.2%)	418.6 (48.7%)	4,685.5 (56.3%)
Low household wealth	3,197.4 (42.8%)	440.2 (51.3%)	3,637.6 (43.7%)
Skilled.birth.attendance			
Skilled.birth.attendance.1	3,225.3 (43.2%)	295.1 (34.4%)	3,520.5 (42.3%)
No.skilled.birth.attendance	4,239 (56.8%)	563.6 (65.6%)	4,802.6 (57.7%)
Sex of the child			
Female	3,828.8 (51.3%)	390.4 (45.5%)	4,219.2 (50.7%)
Male	3,635.5 (48.7%)	468.4 (54.5%)	4,103.9 (49.3%)
Birth order and interval			
First birth	1,396.8 (18.7%)	190.9 (22.2%)	1,587.7 (19.1%)
2-4 short interval	774.3 (10.4%)	110.1 (12.8%)	884.4 (10.6%)
2-4 long interval	2,603.8 (34.9%)	216.6 (25.2%)	2,820.3 (33.9%)
5+ short interval	696.6 (9.3%)	150.6 (17.5%)	847.2 (10.2%)
5+ long interval	1,992.8 (26.7%)	190.7 (22.2%)	2,183.5 (26.2%)
Mother's age at birth			
20 or more	4,256.1 (57.0%)	472.6 (55.0%)	4,728.8 (56.8%)
Less than 20	3,208.1 (43.0%)	386.2 (45.0%)	3,594.3 (43.2%)
Type of residence			
Urban	2,932.1 (39.3%)	263.6 (30.7%)	3,195.7 (38.4%)
Rural	4,532.2 (60.7%)	595.2 (69.3%)	5,127.4 (61.6%)
Mother's education			
Any education	3,342.9 (44.8%)	278.8 (32.5%)	3,621.8 (43.5%)
No education	4,121.4 (55.2%)	579.9 (67.5%)	4,701.3 (56.5%)
Father's education			

Any education	5,490 (73.6%)	589.1 (68.6%)	6,079.2 (73.0%)
No education	1,974.3 (26.4%)	269.7 (31.4%)	2,243.9 (27.0%)
Mother's occupation			
Other	1,610.2 (21.6%)	135.1 (15.7%)	1,745.4 (21.0%)
Household, unskilled, not working	1,757.2 (23.5%)	172.1 (20.0%)	1,929.4 (23.2%)
Agriculture	4,096.8 (54.9%)	551.6 (64.2%)	4,648.4 (55.8%)
Father's occupation			
Other	2,810.3 (37.6%)	242.1 (28.2%)	3,052.4 (36.7%)
Household, unskilled, not working	961.3 (12.9%)	108.5 (12.6%)	1,069.8 (12.9%)
Agriculture	3,692.7 (49.5%)	508.1 (59.2%)	4,200.9 (50.5%)

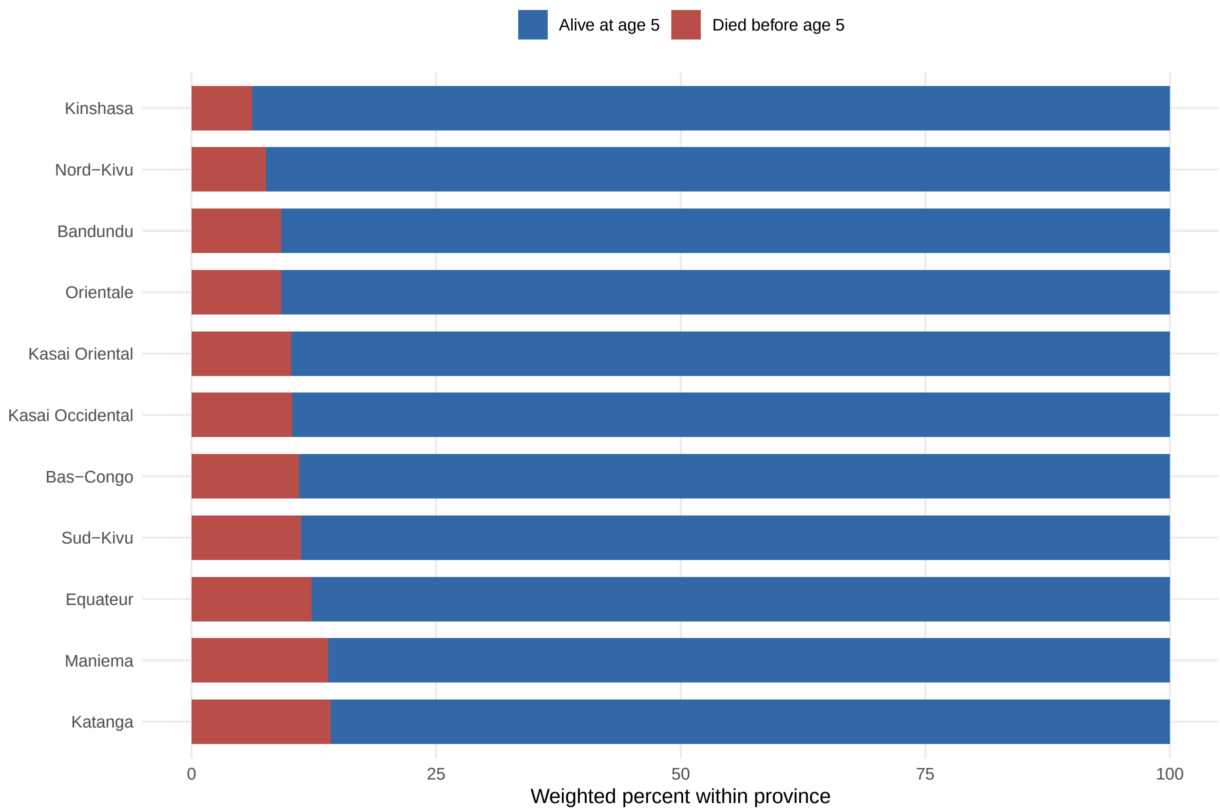

Figure 12: Weighted province distribution by under-five mortality status.

5.1.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 3: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.908	0.275	-10.563	<0.001	-3.448	-2.369
Household wealth	0.068	0.125	0.544	0.5865	-0.177	0.314

Skilled birth attendance	0.198	0.111	1.783	0.0746	-0.020	0.416
Sex of the child	0.242	0.105	2.294	0.0218	0.035	0.449
Birth order and interval: 2-4 short interval	-0.004	0.186	-0.020	0.9838	-0.367	0.360
Birth order and interval: 2-4 long interval	-0.509	0.169	-3.013	0.0026	-0.840	-0.178
Birth order and interval: 5+ short interval	0.420	0.210	1.995	0.0461	0.007	0.832
Birth order and interval: 5+ long interval	-0.396	0.211	-1.882	0.0599	-0.809	0.017
Mother's age at birth	0.033	0.143	0.229	0.8186	-0.247	0.312
Type of residence	0.090	0.154	0.584	0.5591	-0.212	0.392
Mother's education	0.336	0.132	2.555	0.0106	0.078	0.594
Father's education	-0.036	0.128	-0.280	0.7798	-0.286	0.214
Mother's occupation: Household, unskilled, not working	-0.018	0.155	-0.114	0.9093	-0.322	0.287
Mother's occupation: Agriculture	0.140	0.163	0.861	0.3895	-0.180	0.460
Father's occupation: Household, unskilled, not working	0.235	0.162	1.450	0.1472	-0.083	0.554
Father's occupation: Agriculture	0.225	0.149	1.509	0.1314	-0.067	0.517
Region: Bas-Congo	0.258	0.243	1.062	0.2883	-0.218	0.734
Region: Equateur	0.250	0.238	1.051	0.2934	-0.216	0.715
Region: Kasai Occidental	0.105	0.228	0.460	0.6452	-0.342	0.553
Region: Kasai Oriental	0.153	0.240	0.635	0.5251	-0.318	0.624
Region: Katanga	0.515	0.228	2.258	0.0240	0.068	0.962
Region: Kinshasa	0.081	0.262	0.310	0.7564	-0.433	0.596
Region: Maniema	0.489	0.224	2.178	0.0294	0.049	0.929
Region: Nord-Kivu	-0.252	0.294	-0.857	0.3915	-0.827	0.324
Region: Orientale	-0.084	0.265	-0.316	0.7522	-0.603	0.435
Region: Sud-Kivu	0.300	0.270	1.111	0.2667	-0.229	0.829

Predictive ranger tuning results

Table 4: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent
Before undersampling	Alive at age 5	7424	7464.30	89.68
Before undersampling	Died before age 5	872	858.80	10.32
After undersampling	Alive at age 5	1274	1289.74	60.03
After undersampling	Died before age 5	872	858.80	39.97

Table 5: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	22	500	0.6118	0.0083	10	1
accuracy	10	200	0.6108	0.0090	10	2
accuracy	16	500	0.6104	0.0096	10	3
accuracy	29	200	0.6090	0.0106	10	4
accuracy	10	500	0.6080	0.0081	10	5
accuracy	22	200	0.6080	0.0098	10	6
accuracy	16	800	0.6076	0.0042	10	7
accuracy	29	800	0.6076	0.0050	10	8
accuracy	10	800	0.6072	0.0032	10	9
accuracy	22	800	0.6071	0.0051	10	10
accuracy	16	200	0.6066	0.0084	10	11
accuracy	29	500	0.6053	0.0087	10	12
f_meas	29	200	0.3888	0.0175	10	1
f_meas	22	200	0.3753	0.0185	10	2
f_meas	16	200	0.3736	0.0146	10	3
f_meas	10	200	0.3655	0.0155	10	4
f_meas	22	500	0.3459	0.0158	10	5
f_meas	29	500	0.3425	0.0152	10	6
f_meas	16	500	0.3222	0.0176	10	7
f_meas	10	500	0.2867	0.0200	10	8
f_meas	29	800	0.2367	0.0157	10	9
f_meas	22	800	0.2266	0.0194	10	10
f_meas	16	800	0.2068	0.0144	10	11
f_meas	10	800	0.1574	0.0167	10	12
roc_auc	10	500	0.6158	0.0097	10	1
roc_auc	10	200	0.6155	0.0111	10	2
roc_auc	16	500	0.6153	0.0102	10	3
roc_auc	22	500	0.6146	0.0100	10	4
roc_auc	29	500	0.6141	0.0099	10	5
roc_auc	16	200	0.6140	0.0104	10	6
roc_auc	22	200	0.6132	0.0105	10	7
roc_auc	29	200	0.6126	0.0103	10	8
roc_auc	10	800	0.6100	0.0101	10	9
roc_auc	29	800	0.6084	0.0100	10	10
roc_auc	16	800	0.6082	0.0102	10	11
roc_auc	22	800	0.6082	0.0104	10	12
sensitivity	29	200	0.3072	0.0162	10	1
sensitivity	22	200	0.2923	0.0188	10	2
sensitivity	16	200	0.2900	0.0143	10	3
sensitivity	10	200	0.2774	0.0152	10	4
sensitivity	22	500	0.2546	0.0159	10	5

sensitivity	29	500	0.2546	0.0149	10	6
sensitivity	16	500	0.2293	0.0152	10	7
sensitivity	10	500	0.1960	0.0161	10	8
sensitivity	29	800	0.1514	0.0123	10	9
sensitivity	22	800	0.1445	0.0149	10	10
sensitivity	16	800	0.1273	0.0103	10	11
sensitivity	10	800	0.0917	0.0109	10	12
spec	10	800	0.9600	0.0056	10	1
spec	16	800	0.9364	0.0088	10	2
spec	22	800	0.9238	0.0115	10	3
spec	29	800	0.9199	0.0089	10	4
spec	10	500	0.8900	0.0102	10	5
spec	16	500	0.8712	0.0131	10	6
spec	22	500	0.8563	0.0142	10	7
spec	29	500	0.8453	0.0147	10	8
spec	10	200	0.8391	0.0148	10	9
spec	22	200	0.8241	0.0151	10	10
spec	16	200	0.8234	0.0131	10	11
spec	29	200	0.8155	0.0130	10	12

Predictive ranger SHAP summary plot

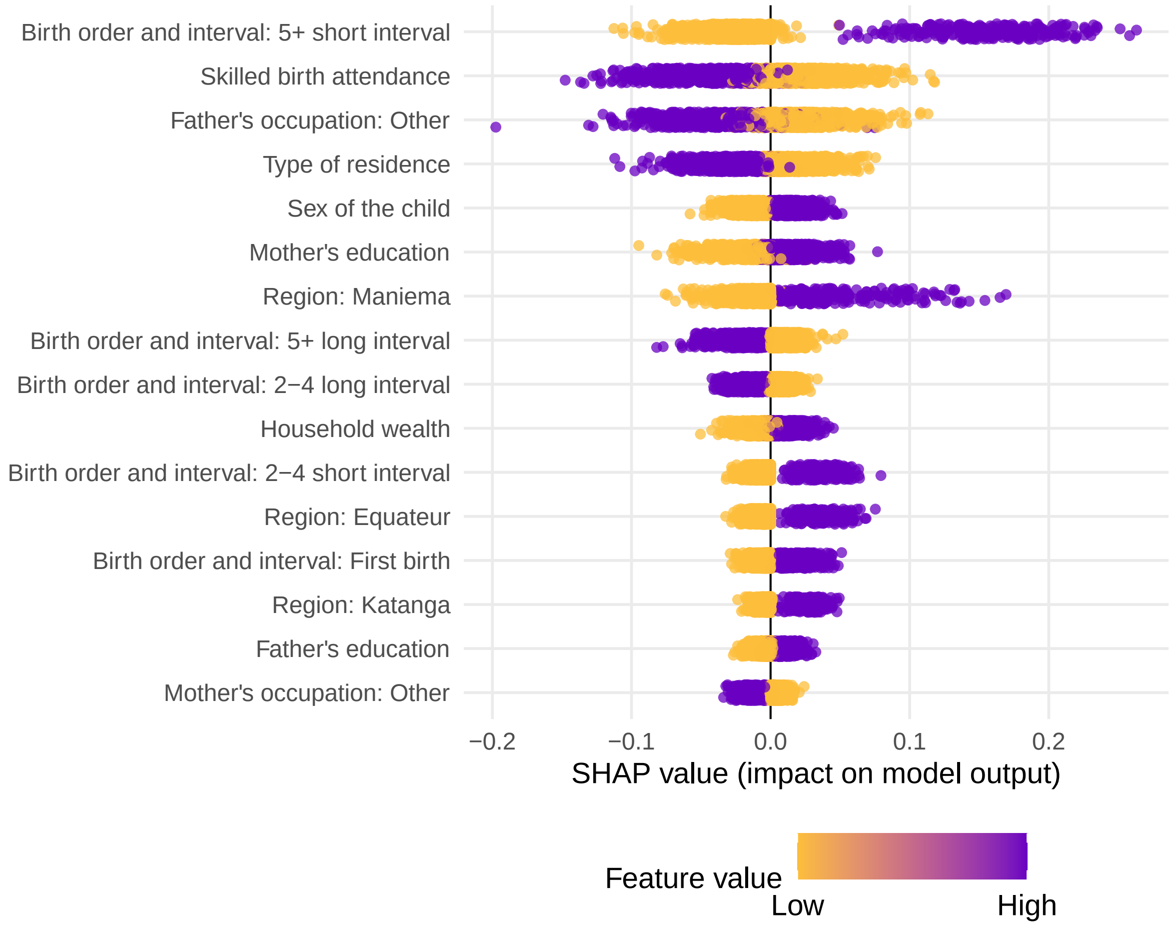

Figure 13: SHAP summary plot for the predictive ranger model.

5.1.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 6: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	100	0.01	10	1e-04	0.3
CIg	500	100	0.01	10	1e-04	0.3
CIc	500	100	0.01	10	1e-04	0.3
L	500	100	0.01	10	1e-04	0.3

Tree fit and selection summaries

Table 7: Tree summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA

CI	Training gain	0.1003	0.0016
CI	Training gain (% root)	87.0688	1.5175
CI	Validation gain	-0.0144	0.0236
CI	Validation gain (% root)	-68.2424	44.7524
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0417	0.0008
CIc	Training gain (% root)	87.5995	1.4760
CIc	Validation gain	-0.0107	0.0134
CIc	Validation gain (% root)	-73.1067	39.0032
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0104	0.0002
CIg	Training gain (% root)	87.5995	1.4760
CIg	Validation gain	-0.0027	0.0034
CIg	Validation gain (% root)	-73.1067	39.0032
L	Root impurity	0.3128	NA
L	Training gain	0.1906	0.0061
L	Training gain (% root)	98.1536	0.2271
L	Validation gain	0.2861	0.1077
L	Validation gain (% root)	70.5254	13.4254

Tree selection metrics

Table 8: Selection metrics for fitted concentration-index trees

Criterion	**Root objective**	**Train gain**	**Train gain/root**	**Validation gain**	**Validation gain/root**	**Terminal nodes**
CI	0.1191	0.1003	0.8707	-0.0144	-0.6824	6.8
CIc	0.0483	0.0417	0.8760	-0.0107	-0.7311	7.3
CIg	0.0121	0.0104	0.8760	-0.0027	-0.7311	7.3
L	0.3128	0.1906	0.9815	0.2861	0.7053	6.8

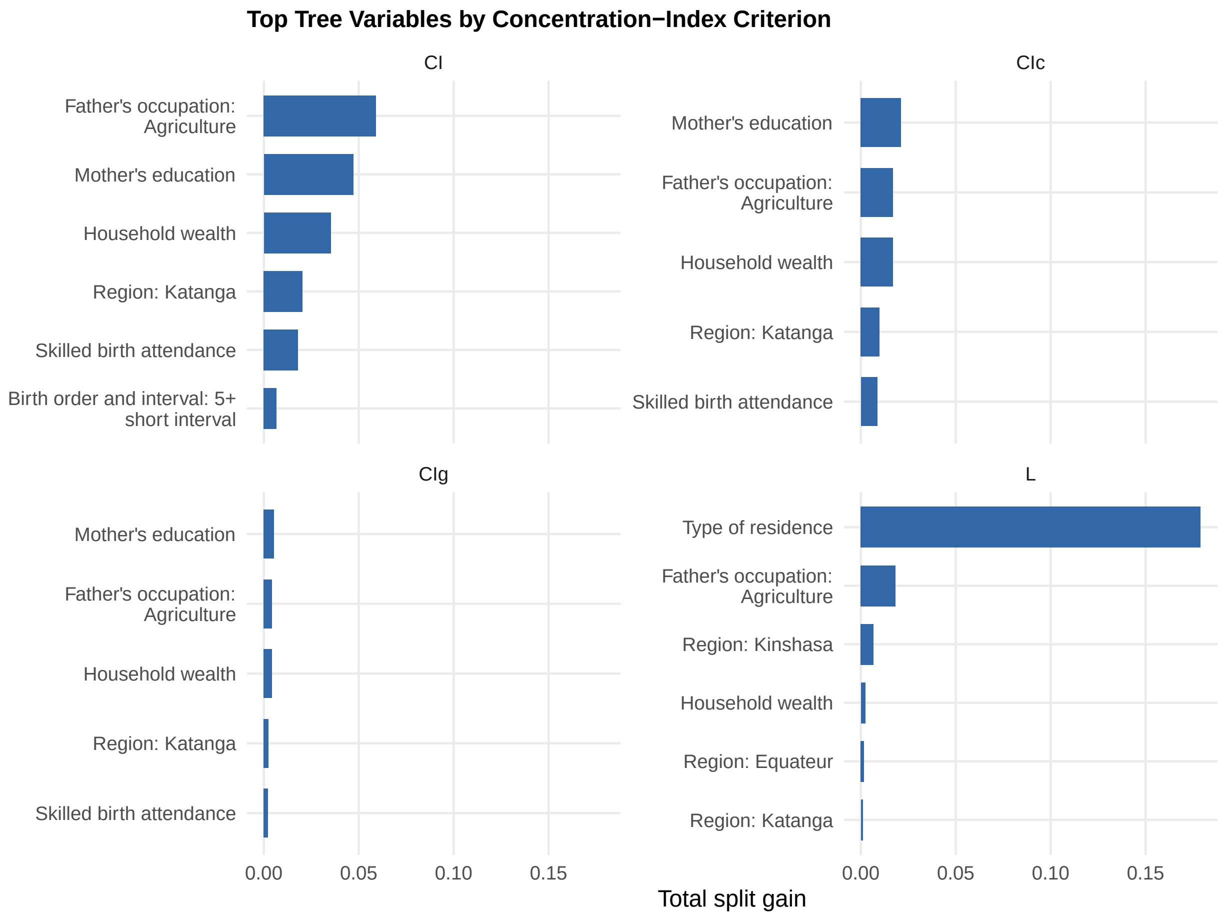

Figure 14: Top tree variables by concentration-index criterion.

5.1.4 Additional Concentration-Index Tree Plots

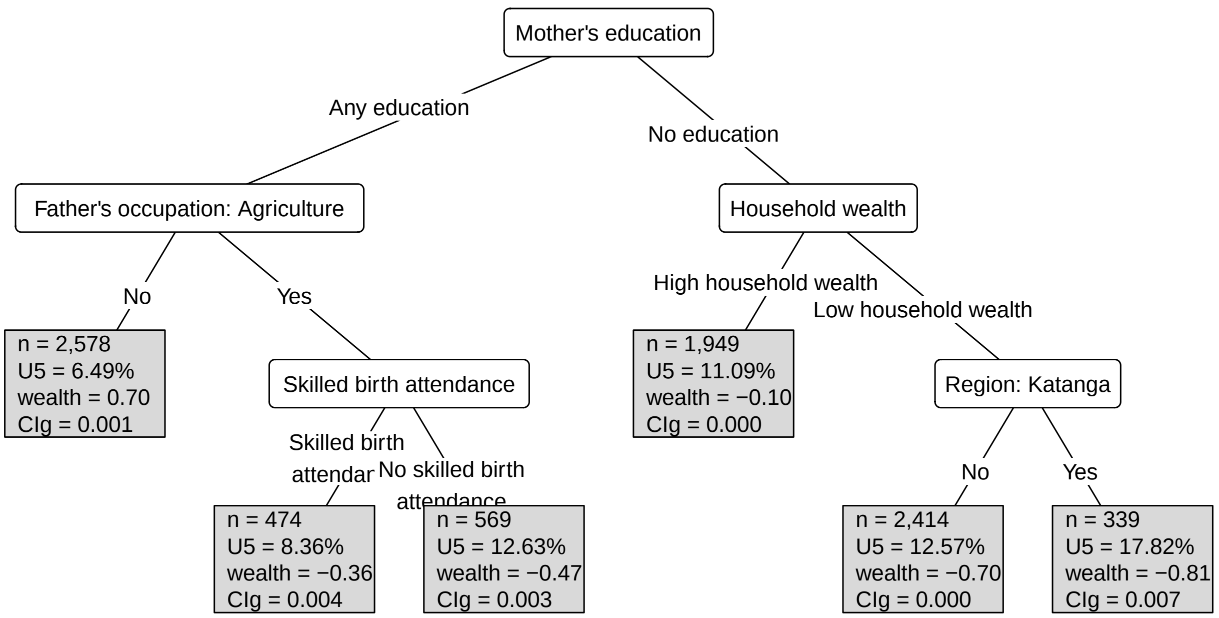

Figure 15: Concentration-index tree fitted with the CIg criterion.

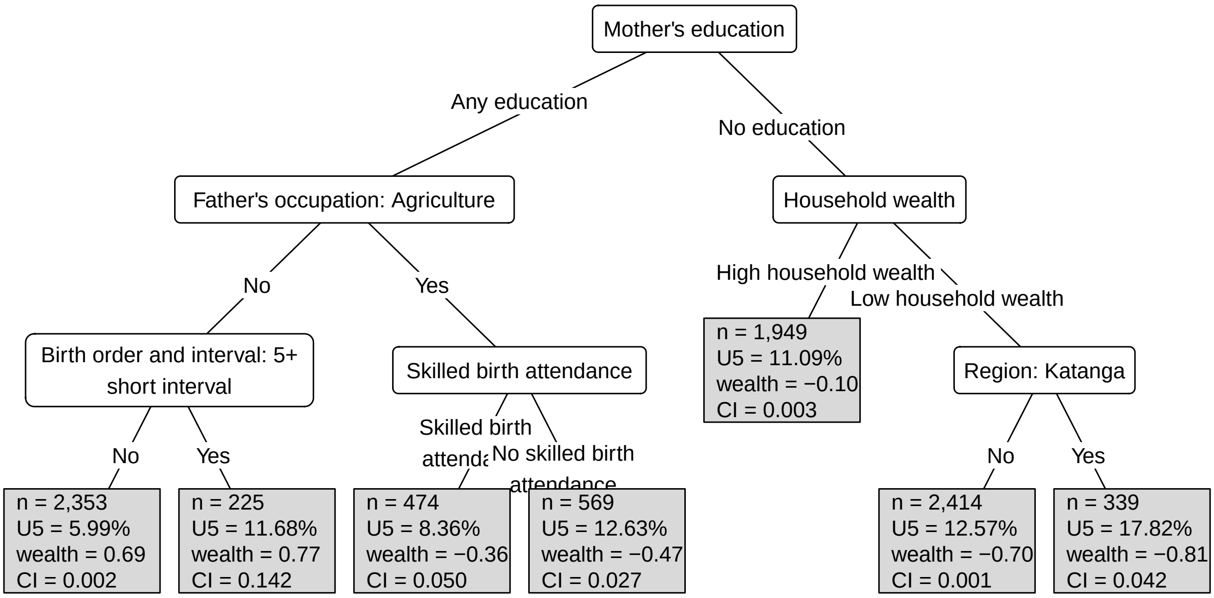

Figure 16: Concentration-index tree fitted with the CI criterion.

5.1.5 Appendix R4: Rpart Comparison Tree Details

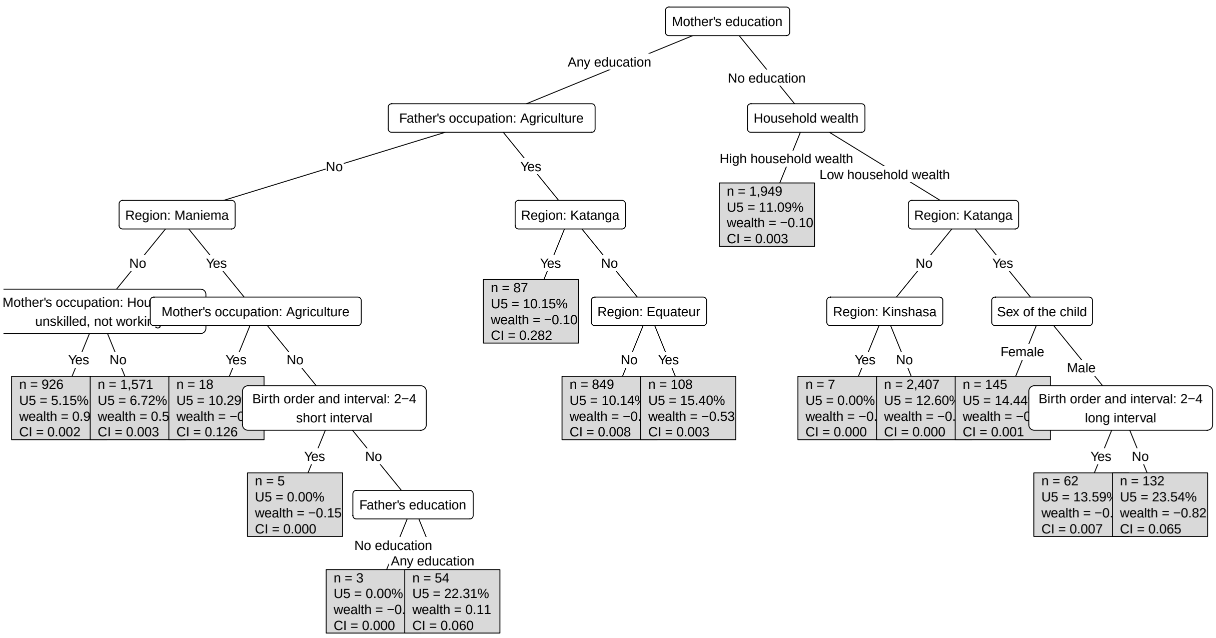

Figure 17: Lecturer rpart comparison tree using absolute CI impurity.

5.1.6 Appendix 5: Concentration-Index Forest Details

Forest tuning and selection details

Table 9: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gai

CI	50	15	500	100	0.01	15	0	0.3
CIc	50	15	500	100	0.01	15	0	0.3
CIg	50	15	500	100	0.01	15	0	0.3
L	50	15	500	100	0.01	15	0	0.3

Forest fit and selection summaries

Table 10: Forest summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.0337	0.0029
CI	Training gain (% root)	29.5891	2.8849
CI	Validation gain	-0.0075	0.0107
CI	Validation gain (% root)	-24.9836	13.0698
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0182	0.0012
CIc	Training gain (% root)	38.5250	2.8199
CIc	Validation gain	-0.0025	0.0054
CIc	Validation gain (% root)	-28.0573	16.5269
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0046	0.0003
CIg	Training gain (% root)	38.6693	2.1250
CIg	Validation gain	-0.0009	0.0015
CIg	Validation gain (% root)	-34.6842	19.5160
L	Root impurity	0.3128	NA
L	Training gain	0.1847	0.0062
L	Training gain (% root)	95.0817	0.1536
L	Validation gain	0.2831	0.1064
L	Validation gain (% root)	69.6578	13.3968

Forest selection metrics

Table 11: Selection metrics for fitted concentration-index forests

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Mean terminal nodes
CI	0.1191	0.0337	0.2959	-0.0075	-0.2498	2.442
CIc	0.0483	0.0182	0.3852	-0.0025	-0.2806	2.864
CIg	0.0121	0.0046	0.3867	-0.0009	-0.3468	2.930
L	0.3128	0.1847	0.9508	0.2831	0.6966	4.070

Forest complexity plots

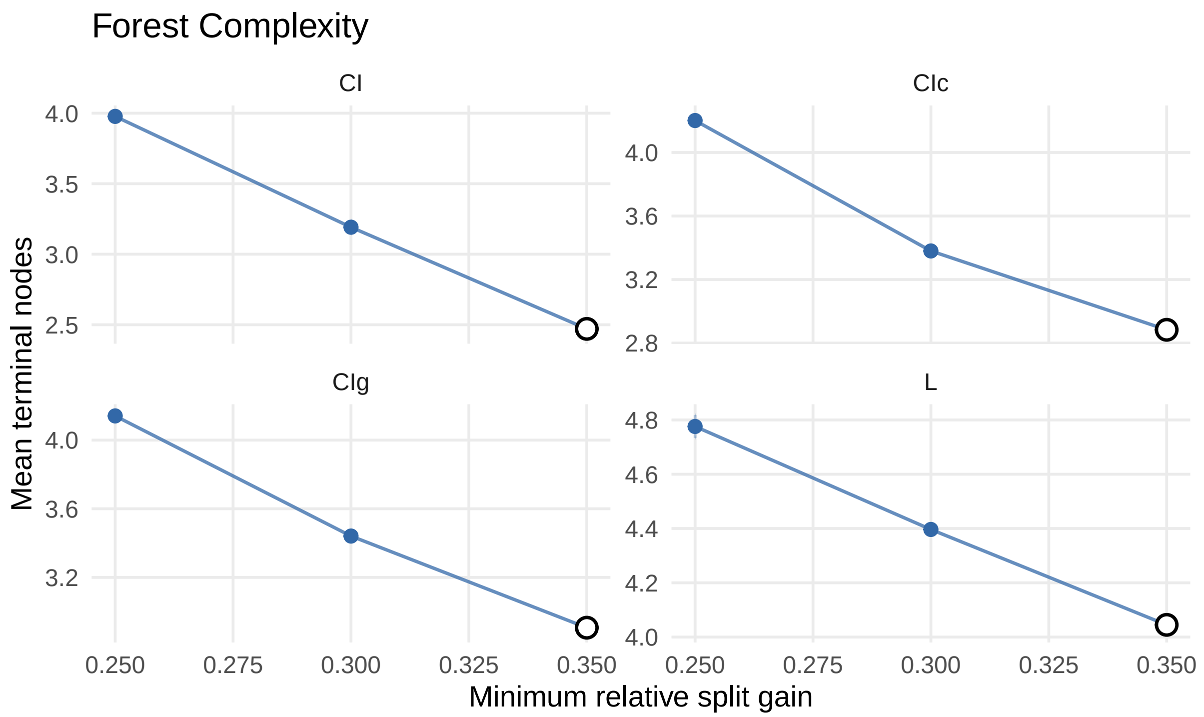

Figure 18: Forest complexity by number of terminal nodes.

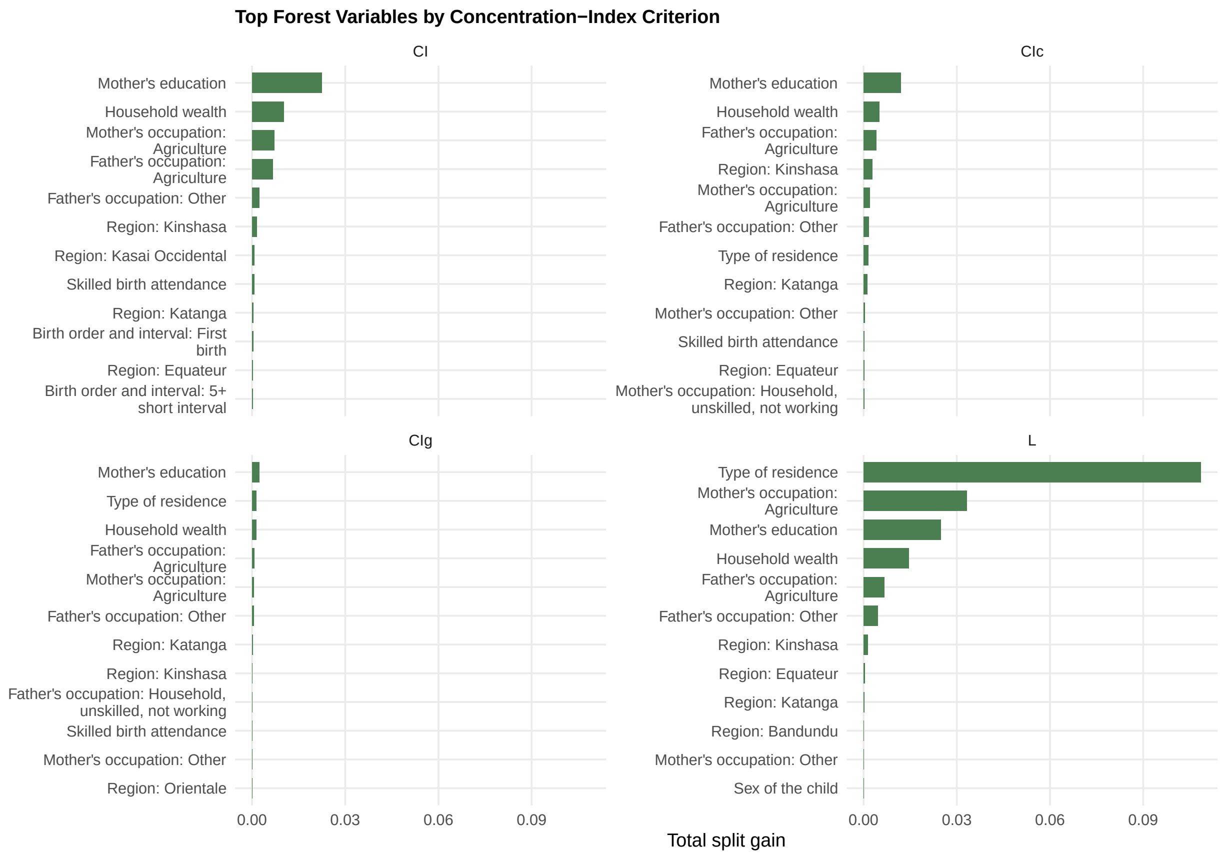

Figure 19: Top forest variables by concentration-index criterion.

Forest Surrogate Trees plots

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

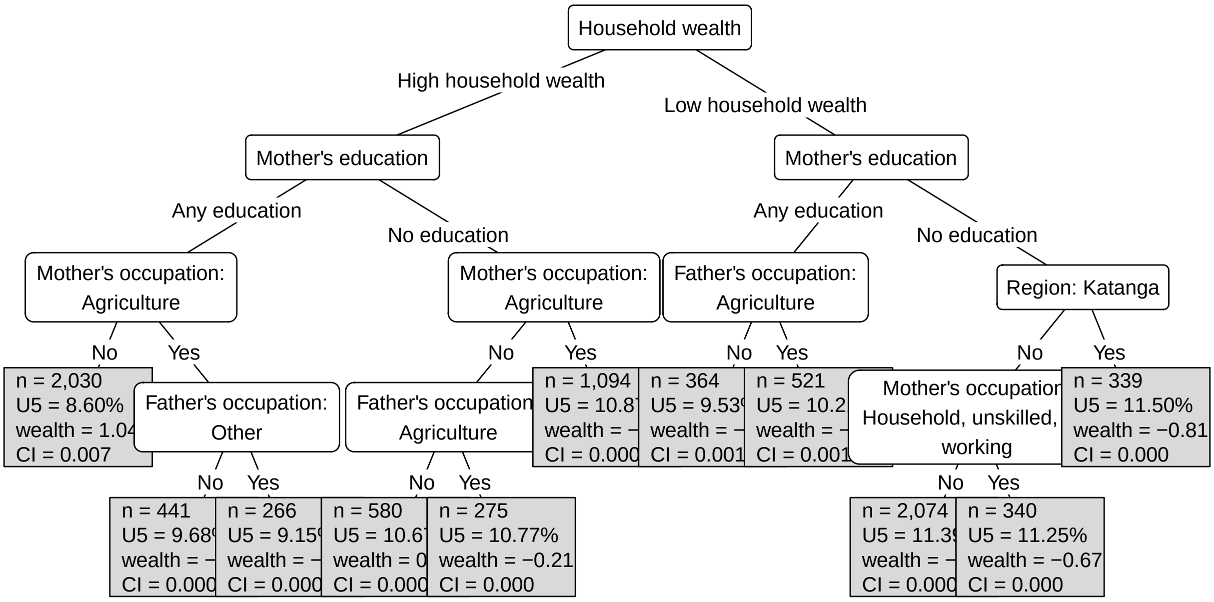

Figure 20: Surrogate tree for the CI concentration-index forest.

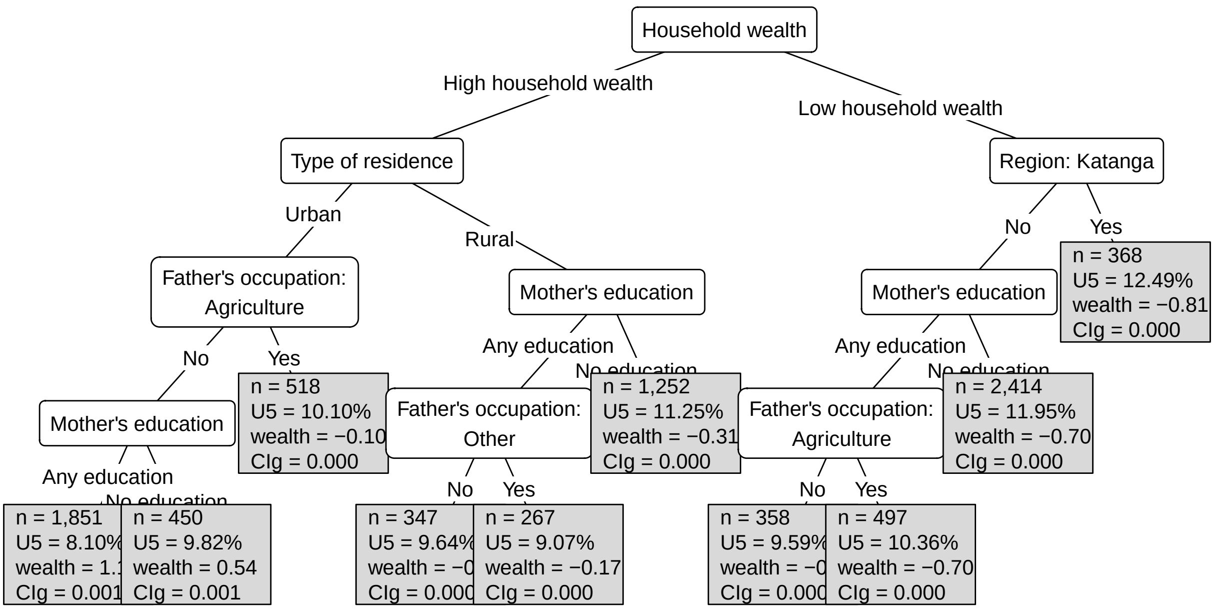

Figure 21: Surrogate tree for the CIg concentration-index forest.

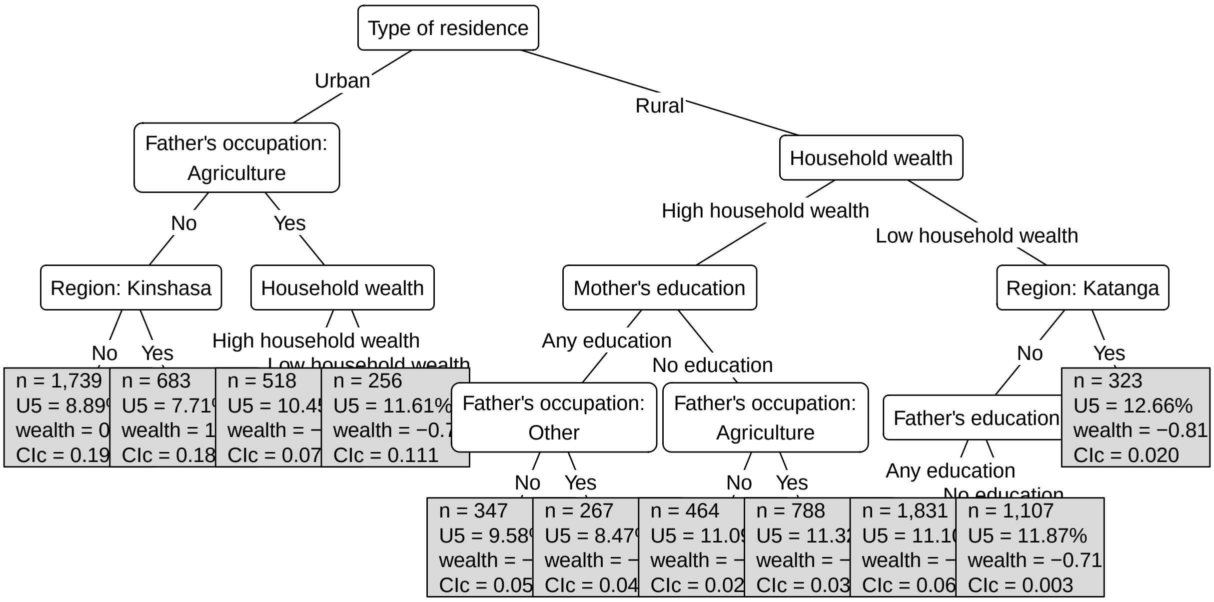

Figure 22: Surrogate tree for the CIc concentration-index forest.

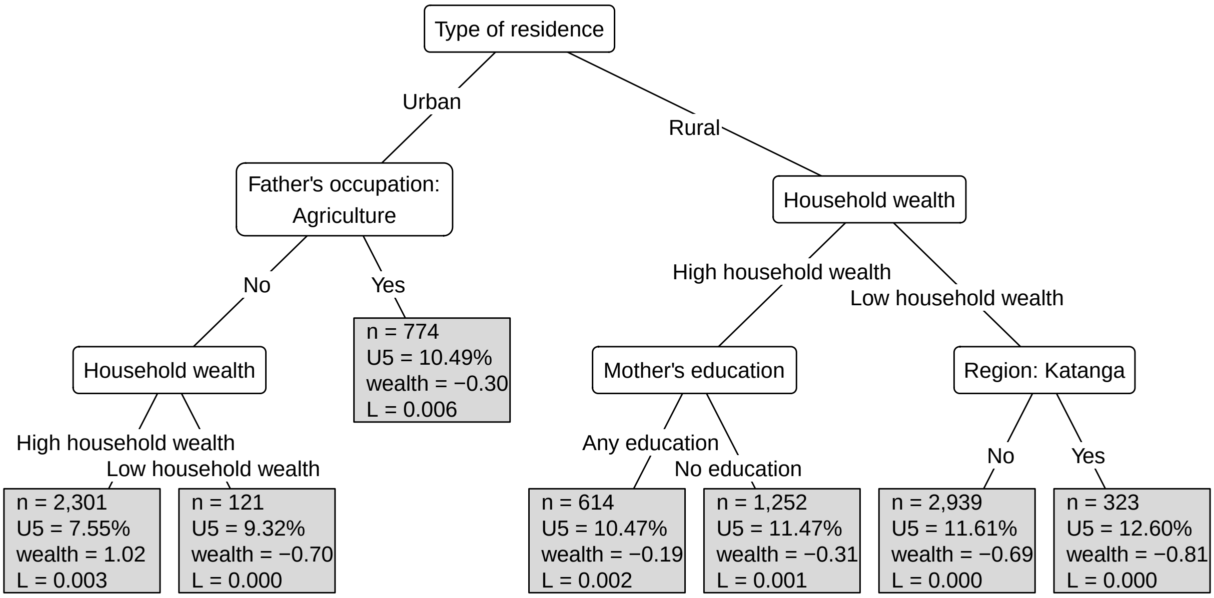

Figure 23: Surrogate tree for the L concentration-index forest.

5.1.7 Appendix S: Simulation Sensitivity Plots

The CIg tree is shown in the main results section. The remaining positive-control simulation trees are shown here to check whether the recovered subgroup pattern is stable across concentration-index criteria.

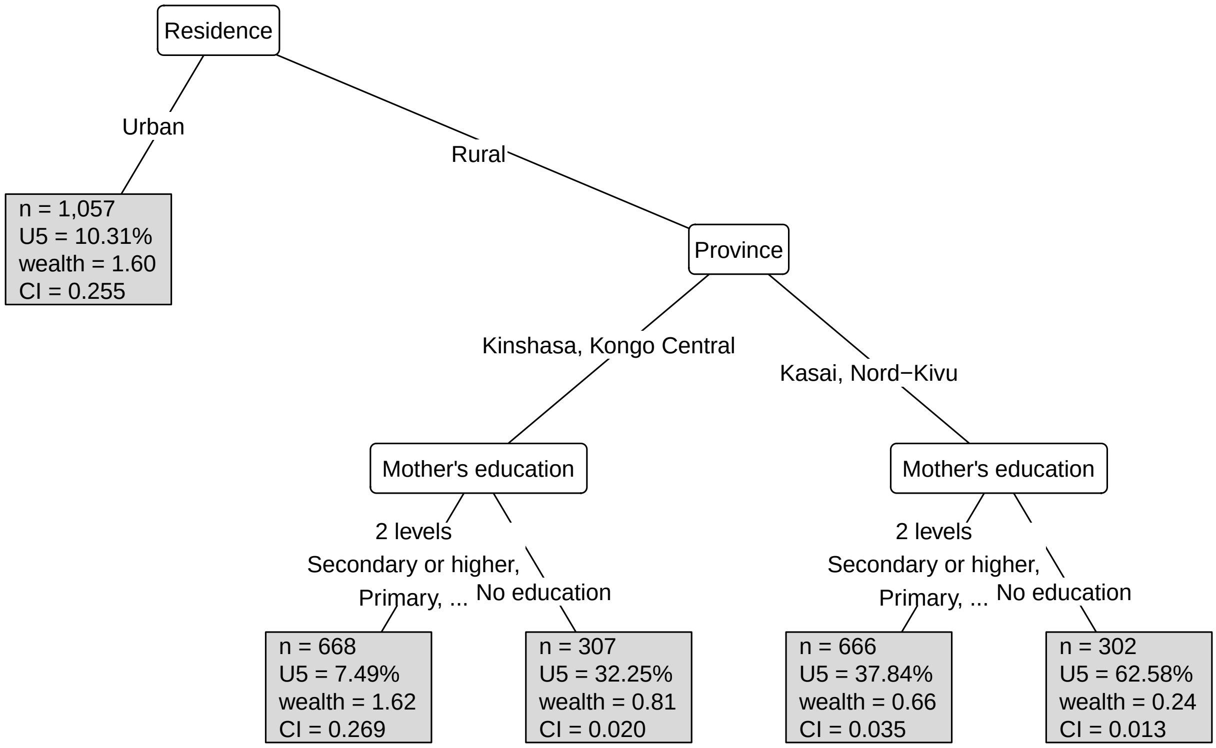

Figure 24: Positive-control simulation tree fitted with the CI criterion.

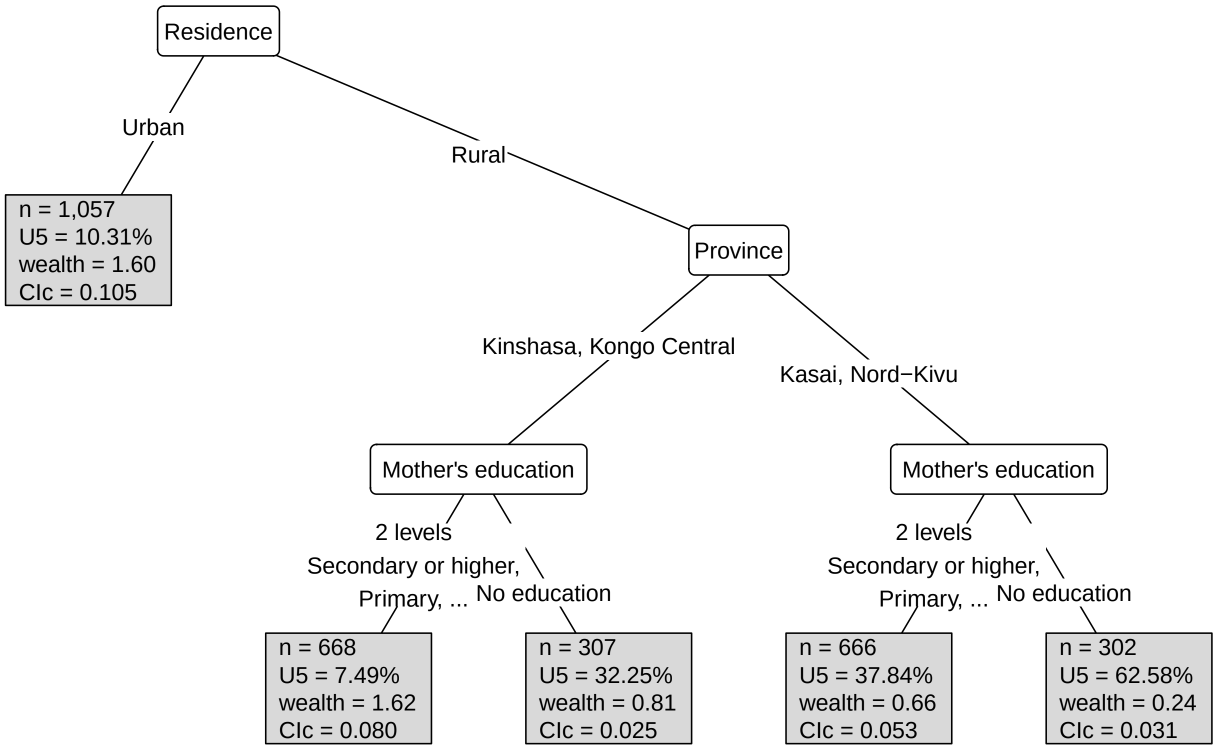

Figure 25: Positive-control simulation tree fitted with the CIc criterion.

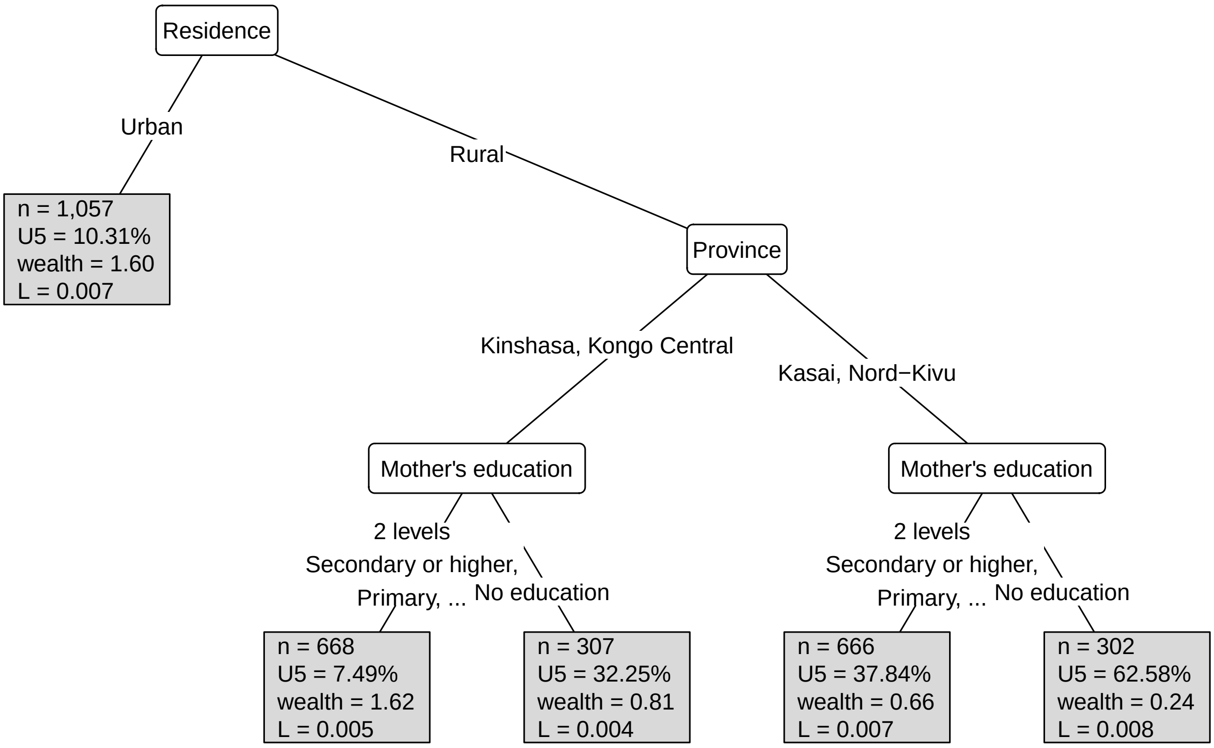

Figure 26: Positive-control simulation tree fitted with the L criterion.

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